#2 · 11.6% of marks

Ch 10 — Forex Exposure & Risk Management

Tested every attempt: swap points / broken date, cross rates & arbitrage, receivable / payable hedges, early delivery & cancellation (FEDAI), exposure theory.

① SWIFT & payment gateways (SM)

- → ***SWIFT*** *(Society for Worldwide Interbank Financial Telecommunication) — **purely a messaging system**; assigns **3-letter currency codes** (USD, GBP, CNY, CAD, AUD, HKD, INR, JPY, NZD, SGD, SEK, CHF, EUR)*
- → ***Why important*** *(S): quick settlement (messaging in seconds) besides validation & documentation · catalyst bringing financial agencies to work together · security & reliability reduce **operational risk** · standardised transactions → efficiency & lower cost · full **backup & recovery***
- → ***Payment gateway*** *— virtual equivalent of physical cash transfer; authenticates & routes payment details securely. **Benefits**: 24×7×365; real-time card authorisation; rapid processing; multiple payment options; encryption & verification; real-time reports; refunds; no complex software for merchants; CA-authenticated secure servers; bulk data collection; multiple host interfaces; simple admin control; customer & merchant trust. **Challenges**: customer may lack account with supporting banks; limited banks; reliability, delivery, limited avenues, lack of trust*

② Accounts, positions, quotes (SM)

Term	SM meaning
Nostro	*Bank's FC account kept in a foreign country in that country's currency — "**our account with you**"*
Vostro	*Account maintained by a foreign bank with a local bank — "**your account with us**"*
Loro	*Domestic bank uses the account of a **third-party bank** that holds a Nostro account, to settle forex transactions*
Exchange position	*Total purchase & sale commitments of a bank, delivered or not*
Cash position	*Balance in Nostro account (credited for purchases, debited for sales). Everything affecting cash position affects exchange position — not vice versa*
American / European term	*American: US$ per unit of foreign currency · European: foreign currency per US$*
Direct / indirect quote	*Direct: home-currency price of 1 unit of foreign (₹48 / $) · Indirect: foreign-currency price of 1 unit of home ($0.0208 / ₹)*
Bid / offer / spread	*Bid = dealer buys; offer = dealer sells; spread = difference*
Cross rate	*Rate for a pair in which **neither** currency is the official currency of the country where quoted*
Pip	*Price Interest Point — smallest unit a quote can change*
Merchant rates	*Rates to customers (vs interbank rates). Post-liberalisation banks set own margins; FEDAI rule SA.8: merchant settlements **rounded to nearest whole rupee**; "card rate" for small walk-in deals*

3 Forward points & broken period (SM)

- → Forward rate is quoted as spot + **forward margin (points)**; user loads margin into spot. Points depend on interest rates & length; the currency with the **lower interest rate is always at a premium**
- → **Premium** (direct quote): **added** to both buying & selling rates · **Discount**: **deducted** from both. Points ascending (bid < ask) → add; descending → subtract
- → **Broken period**: forward points assumed to "grow" **uniformly** — interpolate proportionately

SM illustration — spot ₹66.2525 / 67.5945; 2-m points 70 / 90; 3-m 160 / 186; need 2 m 15 d (1 point = 0.0001)		
Points for 15 days = (3-m − 2-m) ÷ 2	45	48
Points for 2 m 15 d	115	138
Forward rate	**66.2640**	**67.6083**
Annual premium on **average** rate = 0.0115 ÷ 66.2583 × 12/2.5	**0.0833%**	0.0980%

Annualised premium = (F − S) ÷ S × 12 ÷ months × 100; when the question says "on average rate", divide by (S + F) ÷ 2 as above.

4 Cross rates & arbitrage

- → Chain like-to-like: A/C bid = A/B bid × B/C bid; ask = ask × ask. Common base: divide bid by ask
- → SM: ₹/$ 62.22 and ¥/$ 102.34 → ₹/¥ = 62.22 ÷ 102.34 = **0.6080**
- → Arbitrage test: buy via one route at ask, sell via the other at bid; gain only if buy-ask < sell-bid

5 Parity theories (SM)

Theory	Formula	SM summary
IRP	$F \div S = (1 + r_D) \div (1 + r_F)$	Forward premium / discount = interest-rate differential; when IRP holds, **covered interest arbitrage** isn't feasible
PPP	$E(S_1) = S_0 \times (1 + I_d) \div (1 + I_f)$	**Absolute** = law of one price; **relative** allows for imperfections (transport, tariffs, quotas)
IFE	Nominal rate = real return + expected inflation	High-interest currencies **depreciate** (high rates reflect expected inflation; also default risk)

SM comparison	Key variables	Summary
IRP	Forward premium vs interest differential	Forward contains premium / discount set by interest differential
PPP	% change in spot vs inflation differential	Spot changes with inflation differential
IFE	% change in spot vs interest differential	Spot changes with interest differential

App-computed checks: IRP — S 83.50, $r_{₹}$ 7%, $r_{\$}$ 5%, 6 m → F = 83.50 × 1.035 ÷ 1.025 = **84.3146**. PPP — 83.50, inflation 6% vs 3% → **85.932**.

6 Forecasting & determination (SM)

→ *Needed for decisions on hedging, short-term financing & investment, capital budgeting, earnings assessment, long-term financing*

→ ***Techniques (4)**: **Technical** (historical data, time-series) · **Fundamental** (economic relationships — subjective, regression, sensitivity) · **Market-based** (current spot / forward rates reflect expectations) · **Mixed** (weighted average of forecasts)*

→ ***Determinants**: major — **inflation** & **interest** rates; others — deficit / surplus on capital / current account, central bank intervention, expectations (**band-wagon** effect), trade barriers, government controls*

7 Forex market & exposures (SM)

→ *Market permits transfer of purchasing power between currencies. **Participants**: non-bank entities, banks, speculators, arbitrageurs, governments*

Exposure	SM meaning
Transaction	*Effect of rate change on obligations existing **before** the change but settled **after** — cash flows from existing contracts*
Translation	*Need to translate foreign subsidiaries' FS into one reporting currency for consolidation (SM example: FC loan retranslated at year-end)*
Economic	*Extent the **economic value** of the firm can decline due to rate changes — alters input costs & output prices → competitive position*

Economic exposure (SM table)	Local revaluation	Devaluation
*Local sales vs foreign competition; exports (local or foreign currency); interest from foreign investments — **inflows***	*Decrease*	*Increase*
*Imports in foreign currency; interest on foreign debt — **outflows***	*Decrease*	*Increase*
Imports denominated in local currency	*Remain the same*	

✍ **Write in exam**

***Transaction vs translation (May 2026, 2 marks):** transaction exposure affects actual cash flows on settlement of existing contracts, while translation exposure only restates FC items on consolidation without a cash flow — hence transaction exposure is more critical.*

8 Internal hedging techniques (SM — 6)

Invoicing in domestic currency	*Seller invoices in own / cost currency; buyer prefers own or stable currency*
Asset & liability management	***Aggressive**: raise inflows in strong / outflows in weak currencies. **Defensive**: match inflows & outflows by currency*
Leading & lagging	*Leading = pay before due date; lagging = delay after due date*
Netting	*Group companies settle inter-affiliate balances for the net amount*
Matching	*Match FC inflows with FC outflows in amount & timing*
Price variation	*Raise selling prices to counter adverse rate change*

9 Money market hedge

SM: agreement to exchange a certain amount of one currency for a fixed amount of another at a particular date. ***Advantages***: fixes future rate (no downside); flexible amount; usable where forwards aren't available. ***Disadvantages***: more complicated than a forward; no benefit from favourable moves.

Receivable	Payable
1. Borrow FC = receivable ÷ (1 + FC borrowing rate × t)	1. Deposit FC = payable ÷ (1 + FC deposit rate × t)
2. Convert at ***spot bid***	2. Buy FC at ***spot ask*** (borrow home currency if needed)
3. Invest home currency at deposit rate	3. Cost = home borrowing × (1 + rate × t)
4. Receivable repays FC loan; compare with forward	4. FC deposit pays supplier; compare with forward

App-computed: receivable USD 1,00,000 in 6 m, USD borrowing 5% p.a., spot bid 83.40, ₹ deposit 7% → borrow USD 97,560.98 → ₹81,36,585 → **₹84,21,366** at 6 m.

10 Forward vs futures (SM currency table)

Feature	Forward	Futures
Amount	Flexible	Standard
Maturity	Any agreed business date	Standard date
Furthest maturity	Open	12 months forward
Currencies	All	Majors
Cross rates	One contract	Usually two contracts
Market-place	Global network	Futures exchanges
Price fluctuation	No daily limit	Daily limit by exchange
Risk	Counterparty	Minimal (margins)
Honouring	Delivery	Mostly reverse transaction
Cash flow	None till maturity	Initial + variation margin
Trading hours	24 hours	4–8 hour sessions

11 Options vs futures (SM)

- → Only the ***writer*** is obliged (options) vs both parties (futures) · premium paid by buyer vs no premium · option buyer's loss restricted, gain unlimited vs unlimited both ways for futures · American option exercisable any time vs futures honoured on the specified date
- → ***Call*** = right to buy FC at a fixed price on / up to a date (importer); ***put*** = right to sell (exporter)

REPEAT R14 · MCQ FAVOURITE

12 Delivery · Cancellation · Extension (SM)

Event	SM treatment
Delivery on due date	At originally agreed rate, irrespective of spot
Early delivery	Bank may accept if customer bears the bank's loss: fixed charges + **swap difference** (loss / gain from offsetting bank's cover) + **interest on outlay of funds** at not less than **prime lending rate** (if inflow, bank may pass interest at term-deposit rate)
Late delivery	Current rate on delivery date — after applying automatic-cancellation provisions
Cancellation on due date	Flat charges + difference between contracted rate and cancellation rate (**reverse** of original) charged / paid
Early cancellation	At the **opposite** rate of the original contract for the date that matches the original maturity (forward rate)
Late cancellation	Automatic-cancellation provisions apply first
Extension on due date	Cancel at spot (as above) and rebook at forward rate for the new period
Extension before due date	Cancel at relevant forward rate and rebook at current forward rate
Automatic cancellation	**FEDAI Rule 6**: overdue contract without instructions on / before due date is automatically cancelled **within 3 working days after maturity**; customer pays the exchange difference but is **not entitled to any profit**

SM Illustration 4 (early delivery, exporter): contract sell US$1,00,000 at 65.40 for 31 Dec; delivered 30 Nov. Bank sells spot 65.22, buys 1-m forward 65.27 + 0.15 = 65.42 → swap loss 0.20 × 1,00,000 = **₹20,000**; interest on outlay 0.18 × 1,00,000 × 18% × 31/365 = **₹275** → net inflow = 65,40,000 − 20,275 = **₹65,19,725**.

⚠ TRAP

Cancellation = the bank's **reverse action**: customer's purchase contract → bank buys back (TT buying); sale contract → bank sells (TT selling). On automatic cancellation the customer bears losses but never gets gains.

13 NDF & rollover (SM)

- **Non-deliverable forward**: cash-settled, short-term forward on a thinly traded / non-convertible currency; P/L = (agreed rate − spot at settlement) × notional
- **Rollover** = spot cancellation + new contract for a later date. **Reasons**: non-receipt of FC from client (exporter); shortage of local currency (importer); non-agreement of payment with clients; non-availability of longer-period forwards

14 Swaps (SM)

- **Currency swap**: spot exchange of principal → continuing exchange of interest → re-exchange of principal on maturity. **Benefits**: considerable cost savings; access currencies that would otherwise command a high premium; diversification of borrowings; treasurers can hedge currency risk
- **Commodity swap**: series of futures-type settlements on a notional amount at predetermined prices / indices — mainly to hedge **oil price** risk
- **Equity swap**: total return on equity / index (dividend + capital) exchanged for fixed or floating interest

15 Popular derivative products (SM)

Forward importer	Counterparty **buys** notional USD against INR at forward rate
Forward exporter	Counterparty **sells** notional USD against INR at forward rate
FX call option	Counterparty (generally importer) has **right to buy** USD at strike
FX put option	Counterparty (generally exporter) has **right to sell** USD at strike
IRS USD floating to fixed	One party pays fixed, other floating (reset periodically) in the same currency on a notional (constant / amortising / accreting)

16 Strategies for exposure management (SM – 4)

Low risk : low reward	**Automatic hedging** in forward market as soon as exposure arises
Low risk : reasonable reward	**Selective hedging** — hedge when forward rates attractive, keep open otherwise
High risk : low reward	Leave all exposures **unhedged** — perhaps the worst
High risk : high reward	**Active trading** — continuous cancellations & re-bookings of forwards

Choice depends on attitude to risk, financial strength, nature of business, vulnerability to adverse movements — no single strategy suits all.

17 Exam method

→ Identify home currency, receivable / payable; exporter uses **bid**, importer **ask**; compare forward / MMH / futures / options / unhedged and **recommend**

18 Correlation — how Ch 10 actually connects

Ch 10 idea	The actual link (app-written cross-reference; the Ch 10 side is SM)
Interest rate parity	**Ch 9**: same no-arbitrage logic as cost of carry. Forward premium/discount % ≈ interest differential; if not → covered interest arbitrage (borrow cheap currency, convert spot, invest, cover forward).
Money market hedge	**Ch 5**: uses money-market borrowing / deposit rates for the exposure period — simple interest, days ÷ 360 or 365 exactly as the question states.
PPP / forecasting	**Ch 11**: foreign project cash flows are converted at forecast rates (PPP / IRP); or discount in foreign currency at foreign rate and convert at spot — same answer only if parity holds.
Currency options / futures	**Ch 9**: identical payoff and margin / MTM mechanics; the 11-row forward vs futures table applies to both chapters.
Currency swaps	**Ch 12**: same comparative-advantage logic as interest rate swaps — each party borrows where it is relatively cheaper and swaps.
Translation of exposure	**FR Ind AS 21**: monetary items at closing rate, difference to P&L; **Ind AS 109**: forward contract is a derivative at fair value.